Abhinav Gorantla

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Education

Arizona State University, *School of Computing and Augmented Intelligence* — *Jan 2026 – Present*
Ph.D. in Computer Science, GPA: 4.0/4.0

- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.

Arizona State University, *School of Computing and Augmented Intelligence* — *Aug 2023 – Dec 2025*
Master of Science in Computer Science, GPA: 4.0/4.0

- **Honors:** With Distinction.
- **Thesis:** *Speeding-up of the Non-Dominated Sorting Genetic Algorithm-II by Selective De-Correlation*
- **Advisor:** Dr. K. Selçuk Candan.

Vellore Institute of Technology, *School of Computer Science and Engineering* — *Jul 2019 – May 2023*
Bachelor of Technology in Computer Science and Engineering

- **GPA:** 8.94/10.0

Publications

The Good, the Bad, and the Ugly of Markov Boundary for Tabular Prediction
Shu Wan, ***Abhinav Gorantla***, Huan Liu, K. Selçuk Candan
Accepted to 35th ACM International Conference on Information and Knowledge Management (CIKM '26), 2026.
Showed that a target variable oracle Markov boundary can substantially improve tabular prediction, but recovering it via causal discovery fails to deliver - motivating prediction-aligned feature selection.
Preprint: arxiv.org/abs/2605.29411

Causal Search for Skylines (CSS): Causally-Informed Selective Data De-Correlation
Pratanu Mandal, ***Abhinav Gorantla***, K. Selçuk Candan, Maria Luisa Sapino
In the Proceedings of the ACM on Management of Data, Volume 4, Issue 3 (SIGMOD 2026)
Proposes a method for leveraging causal structure to selectively de-correlate data, improving efficiency of Skyline retrieval in relational databases.

DOI: 10.1145/3802026
Code: github.com/EmitLab/Causal-Search-for-Skylines

CausalBench-ER: Causally-Informed Explanations and Recommendations for Reproducible Benchmarking
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu, K. Selçuk Candan
In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), 2025.
Extends CausalBench with a recommendation engine that suggests appropriate causal analysis pipelines based on dataset, model and hyperparameter characteristics.

DOI: 10.1145/3746252.3761606

Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning *[Best Demo Award]*
Ahmet Kapkiç, Pratanu Mandal, Shu Wan, Paras Sheth, ***Abhinav Gorantla***, Yoonhyuk Choi, Huan Liu, K. Selçuk Candan
In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), 2024.
An open benchmarking platform for causal learning and machine learning.

DOI: 10.1145/3627673.3679218

Pre-Prints

VoS: Variate Ordering Strategies for Skyline Query Optimization
Abhinav Gorantla, Pratanu Mandal, K. Selçuk Candan, Maria Luisa Sapino
Skyline query efficiency depends heavily on data characteristics, and in this paper we argue that algorithms should optimize per-attribute dominance checks (not just per-tuple ones), proposing variate-ordering strategies to cut redundant attribute comparisons.
Preprint: arxiv.org/abs/2608.26464

Tutorials

Spatio-Causal Modeling and Applications
Fahim Tasneema Azad, Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Maria Luisa Sapino, Huan Liu,
K. Selçuk Candan
Accepted to tutorial track at the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026), 2026.
Website: spatial.causalbench.org

CausalBench: Causal Learning Research Streamlined
Ahmet Kapkiç, Pratanu Mandal, ***Abhinav Gorantla***, Shu Wan, Ertugrul Çoban, Paras Sheth, Huan Liu,
K. Selçuk Candan
Tutorial at the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025), 2025.
DOI: 10.1145/3711896.3737598
Website: tutorial.causalbench.org

Research and Professional Experience

Graduate Research Assistant — *Tempe, AZ*
EMIT Lab, Arizona State University — *August 2024 - (present)*
- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning, Multi-Objective Optimization.
- Working on causally informed training methods for Prior Fitted Networks (PFNs).
- Developing an optimized algorithm for efficient Skyline retrieval in relational database systems by leveraging causal structure in the data.
- Conducting research on optimizing the selection process in multi-objective evolutionary algorithms, specifically Pareto-based methods like NSGA and SPEA.
- Collaborating with researchers at CASCADE Lab to maintain and improve `causalbench.org`, a platform dedicated to benchmark causal learning algorithms.
- Collaborated with researchers at CASCADE Lab in developing a novel causally informed recommendation system for `causalbench.org`.

Graduate Services Assistant — *Tempe, AZ*
EMIT Lab, Arizona State University — *March 2024 - August 2024*
- **Advisor:** Dr. K. Selçuk Candan.
- **Research areas:** Causally informed Machine Learning.
- Supported CASCADE Lab researchers in developing the causalbench-asu Python package and website, establishing an end-to-end benchmarking solution for the causal machine learning community.
- Served as a full stack developer on the CausalBench project, contributing to a comprehensive framework for benchmarking causal machine learning algorithms.

Software Development Engineer Intern — *Bengaluru, India*
Webknot Technologies Pvt. Ltd. — *April 2022 - June 2023*
- Revamped API endpoints within the Palette project, achieving a notable 30% reduction in response times.
- Engineered a custom plugin for Sisense BI software, enabling the seamless display of geojson data on a GeoJSON layer atop maps rendered via DeckGL.

- Fine-tuned data flow for the DeckGL plugin within Sisense by elevating the efficiency of JAQL queries, ensuring a smoother and more responsive user experience.

Projects

Enhancing Diversity in the LLM Modulo Framework through Multi-Response Generation

- Developed the Diversified LLM Modulo framework to address looping and redundancy in the LLM Modulo framework.
- Improved the performance of the LLM Modulo Framework on Planning tasks. Tested my framework on the Google Deepmind Natural Plan benchmark (paper) and achieved a performance improvement of 300% by increasing the diversity of LLM (Large Language Model) Responses.
- Tools Used: Python, OpenAI API.

Research Publications Analysis tool

- Architected and developed a research publication analysis web application for ASU, integrating AWS S3, SageMaker, and SES. This tool leverages SCOPUS APIs to fetch research papers and perform text analysis of abstracts.
- Utilized LLM (GPT-4) for advanced feature extraction from scraped text.
- Optimized back-end system performance by reducing server response time by 80% and production costs by 40% through the strategic integration of AWS SageMaker.
- Tools Used: ReactJS, NodeJS, Python-FastAPI, RabbitMQ, MongoDB, AWS S3, AWS Sagemaker, OpenAI API.

Teaching Experience

ASU SCience and ENgineering Experience (SCENE) — Arizona State University, Spring 2026
Co-mentored a SCENE high school student on event detection in multivariate time-series data; supported experimental design and preparation for AZSEF.

Graduate Teaching Assistant — Arizona State University, Spring 2025
CSE 510 Database Management Systems Implementation

Graduate Teaching Assistant — Arizona State University, Fall 2024
CSE 515 Multimedia and Web Databases

Professional Service

Program Committee Member (Reviewer)
ACM Conference on Information and Knowledge Management (CIKM) 2026

Student Volunteer
ACM Knowledge Discovery and Data Mining (KDD) 2025 Conference

Awards

University Graduate Fellowship 2026, School of Computing and AI
Arizona State University

Best Demo Award, CIKM 2024 For *Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning*

Skills

Programming: Python, C++, C, Java, TypeScript.

Machine Learning: PyTorch, NumPy, Pandas, Scikit-learn.

Databases Technologies: MongoDB, SQL, SQLAlchemy.

Cloud Technologies: AWS S3, AWS EC2, AWS Sagemaker, AWS Lambda, Google Firestore, Google Firebase

Functions.

Web Technologies: NodeJS (family), ReactJS, FastAPI.

Other: Git, Tensorflow, RabbitMQ.

Leadership

Vice Chairperson *Vellore, India*
THEP Journalism Club, Vellore Institute of Technology

- Established and sustained a robust MERN stack application as the cornerstone of the club's online newsletter platform.
- Pioneered the creation of a podcast for our club, achieving an impressive audience of nearly 800 listeners during its inaugural month.

Editor-in-Chief *Vellore, India*
Fifth Pillar NGO, Vellore Institute of Technology

- Oversaw and directed a 50-member editorial team, responsible for the editing and publication of articles on our blog website.
- Innovated by introducing new content formats like "Law Talks," resulting in a remarkable 50% expansion in the club's social media and community outreach.

Professional Membership and Affiliations

- Association for Computing Machinery (ACM)
  - Special Interest Groups - SIGMOD, SIGIR, SIGWEB.
- American Association for the Advancement of Science (AAAS)

[0]References available upon request.